

Docker

- is a virtualization software that makes developing and deploying applications much easier
- packages application with necessary dependencies, configurations, system tools and runtime
- its like winrar but for projects
- container, instead of downloading all items to get a project started or having os issues, everything is bundled up in the docker item
- docker artifact has everything needed, no configs needed on server
- run docker commands to fetch and run the artifacts
- difference between vm and docker is that vm's also virtualize the OS Kernel, compared to docker only visualizing only the applications layer
- linux based docker images cannot use windows kernels

Docker Images & Containers

- docker image is an executable application artifact, source code, environment configuration
 - application - JS app
 - any services - node, npm
 - OS Layer - Linux
- docker container is what actually starts the application
 - a running instance of an image
 - when the container environment is created
 - can run multiple containers from 1 image

Docker Registries & Repositories

- a storage and distribution system for docker images
- official images available from applications like Redis, Mongo, Postgres

- Docker Hub - largest docker hosted registry - <https://hub.docker.com/>
- there are both public and private repositories
 - amazon has some within AWS ECR
- repository - a collection of related images, same name but different versions

Image Versioning

- new software versions, will also create different docker image versions, which in docker world is called tags

Container Port vs Host Port

- application inside container runs in isolated docker network - need to expose container port to host (port binding) - best practice is to have host port be the same as container port

Dockerfile - Build Instructions

- text document that contains commands to assemble an image, docker will build an image by reading those instructions

Docker Desktop



